

Zibo Liu

Public Academic CV · Acoustics and Soundscape Engineering

Affiliated Researcher, Department of Engineering Mechanics, KTH Royal Institute of Technology
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RESEARCH PROFILE

Zibo Liu is an acoustics researcher working across physical acoustics, acoustic metamaterials, computational design, and human centred soundscape engineering. His research connects wave physics modelling with experimental and perceptual approaches to sound.

ACADEMIC APPOINTMENTS

2025–present	Affiliated Researcher KTH Royal Institute of Technology
2024–2025	Affiliate Academic University College London
2021–2023	Postdoctoral Researcher Tsinghua University
2022–2023	Visiting Postdoctoral Researcher KTH Royal Institute of Technology
2019–2021	Assistant Researcher Institute of Acoustics, Chinese Academy of Sciences

EDUCATION

2019	PhD, Engineering Acoustics KTH Royal Institute of Technology Thesis: Design of Soundproof Panels via Metamaterial Concept
2014	MSc, Theoretical Acoustics National University of Defense Technology
2011	BEng, Applied Mechanics Beijing Institute of Technology

SELECTED RESEARCH FUNDING

- **SMILE**, Principal Investigator, Swedish Research Council, 2026–2029 (grant 2025-06253).
- **SMILE City: A Pre-study**, Co-Principal Investigator, Stockholm Trio–UCL Seed Fund, 2025–2026.
- **Phonon contribution to energy dissipation in nanofriction**, Principal Investigator, National Natural Science Foundation of China, 2022–2024.
- **Superlubrication enabled by superlattices**, Principal Investigator, China Postdoctoral Science Foundation, 2022–2024.

SELECTED PUBLICATIONS

Liu, Z., Rumpler, R., Feng, L. (2021). “Locally resonant metamaterial curved double wall to improve sound insulation at the ring frequency and mass-spring-mass resonance.” *Mechanical Systems and Signal Processing*, 149, 107179.

Liu, Z., Rumpler, R., Sun, H., Li, Q., Liu, D., Yu, W. (2022). “Improving sound insulation near ring and coincidence frequencies of cylindrical sandwich shells.” *International Journal of Mechanical Sciences*, 235, 107661.

Liu, Z., Rumpler, R., Feng, L. (2019). “Investigation of the sound transmission through a locally resonant metamaterial cylindrical shell in the ring frequency region.” *Journal of Applied Physics*, 125, 115105.

Liu, Z., Rumpler, R., Feng, L. (2018). “Broadband locally resonant metamaterial sandwich plate for improved noise insulation in the coincidence region.” *Composite Structures*, 200, 165–172.

Dai, H., Wang, Y., Liu, Z., Liu, Y., Guo, Y., Liu, D. (2025). “Interlayer phononic energy dissipation in the friction of graphene layers.” *Science Advances*, 11, eadu2880.

Xia, Y., Chen, Y., Liu, Z., Guo, Q. (2026). “Two-dimensional chiral Weyl points in an interlaced Kagome layer.” *Optics Letters*, 51, 2888–2891.

SELECTED ACADEMIC ACTIVITIES

- Co-supervision of doctoral and master's research in acoustics, soundscape, and phonon engineering.
- Teaching contributions in building acoustics, nanomanufacturing, interface science, and phonon engineering.
- Member of the Acoustical Society of America and the International Institute of Acoustics and Vibration.